

Experiment 11 — Orchestration using Docker Compose & Docker Swarm

Objective

To understand container orchestration and deploy a multi-container application using Docker Swarm. The experiment demonstrates scaling, load balancing, and self-healing capabilities compared with Docker Compose.

Theory

What is Container Orchestration?

Container orchestration is the **automatic management of containers**.

It handles tasks such as:

- scaling containers
- restarting failed containers
- distributing traffic across containers
- running services across multiple machines

An analogy is a **restaurant manager** who ensures enough staff are working, replaces unavailable staff, and distributes customers efficiently.

Progression of Container Management Tools

docker run → Docker Compose → Docker Swarm → Kubernetes

Tool	Purpose
docker run	run a single container
Docker Compose	run multiple containers together
Docker Swarm	orchestrate containers across nodes
Kubernetes	advanced orchestration platform

This lab focuses on **moving from Compose to Swarm orchestration**.

Prerequisites

- Docker installed
 - Docker Swarm enabled
 - WordPress + MySQL Compose file from Experiment 6
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Docker Compose File

```
version: '3.9'

services:

  db:
    image: mysql:5.7
    container_name: wordpress_db
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: rootpass
      MYSQL_DATABASE: wordpress
      MYSQL_USER: wpuser
      MYSQL_PASSWORD: wppass
    volumes:
      - db_data:/var/lib/mysql

  wordpress:
    image: wordpress:latest
    container_name: wordpress_app
    depends_on:
      - db
    ports:
      - "8080:80"
    restart: always
    environment:
      WORDPRESS_DB_HOST: db:3306
      WORDPRESS_DB_USER: wpuser
      WORDPRESS_DB_PASSWORD: wppass
      WORDPRESS_DB_NAME: wordpress
    volumes:
      - wp_data:/var/www/html

volumes:
  db_data:
  wp_data:
```

Task 1 — Stop Existing Containers

First ensure no containers are running.

```
docker compose down -v
docker ps
```

Expected result: no containers running.

Task 2 — Initialize Docker Swarm

Enable Swarm mode:

```
abhinav_pandey@LUCIF3R: ~/exp6$ docker swarm init --advertise-addr 172.22.87.136
Swarm initialized: current node (v6zzfufq8onqnlapf8k1oieez) is now a manager.

To add a worker to this swarm, run the following command:

    docker swarm join --token SWMTKN-1-lwnm13ttv0ooo7p2yy41blofcke95gv3ex0wh3eo55kagh9e9-6qsd51g19v34ta54kbinuu5p9 172.22.87.136:2377

To add a manager to this swarm, run 'docker swarm join-token manager' and follow the instructions.
```

This command:

- enables swarm mode
- makes the current machine the **manager node**
- generates a join token for worker nodes

Verify swarm status:

```
abhinav_chaula@LUCIF3R: ~/exp6$ docker node ls
ID                                HOSTNAME    STATUS    AVAILABILITY    MANAGER STATUS    ENGINE VERSION
v6zzfufq8onqnlapf8k1oieez *      LUCIF3R    Ready    Active           Leader             29.4.0
```

Expected output shows the node status as **Ready / Leader**. :contentReference[oaicite:4]{index=4}

Task 3 — Deploy Application as a Stack

Deploy the compose file as a **stack**.

```
abhinav_pandey@LUCIF3R: ~/exp6$ docker stack deploy -c docker-compose.yml wpstack
Ignoring unsupported options: restart

Ignoring deprecated options:

container_name: Setting the container name is not supported.

Since --detach=false was not specified, tasks will be created in the background.
In a future release, --detach=false will become the default.
Creating network wpstack_default
Creating service wpstack_db
Creating service wpstack_wordpress
```

Explanation:

Command	Meaning
docker stack deploy	deploy application in swarm
-c docker-compose.yml	specify compose file
wpstack	stack name

Behind the scenes Swarm:

1. reads the compose file
2. creates services
3. manages containers automatically.

Task 4 — Verify Deployment

List services:

```
abhinav_chaula@LUCIF3R: ~/exp6$ docker service ls
```

ID	NAME	MODE	REPLICAS	IMAGE	PORTS
s0shrokwgbc1	wpstack_db	replicated	1/1	mysql:5.7	
t7fl7sa6r0t5	wpstack_wordpress	replicated	0/1	wordpress:latest	*:8080->80/tcp

Example output:

```
ID NAME MODE REPLICAS IMAGE
xxxx wpstack_db replicated 1/1 mysql:5.7
xxxx wpstack_wordpress replicated 1/1 wordpress:latest
```

Inspect running tasks:

```
abhinav_chamta@LUCIF3R: ~/exp6$ docker service ps wpstack_wordpress
```

ID	NAME	IMAGE	NODE	DESIRED STATE	CURRENT STATE	ERROR	PORTS
s88w74v0994n	wpstack_wordpress.1	wordpress:latest	LUCIF3R	Running	Running 3 seconds ago		

Check containers:

```
abhinav_pandey@LUCIF3R: ~/exp6$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
2faf08231f4a	wordpress:latest	"docker-entrypoint.s..."	24 seconds ago	Up 23 seconds	80/tcp	wpstack_wordpress.1.s8
8w74v0994ntwa4	mysql:5.7	"docker-entrypoint.s..."	About a minute ago	Up About a minute	3306/tcp, 33060/tcp	wpstack_db.1.twogu8ihb

Containers now have names like:

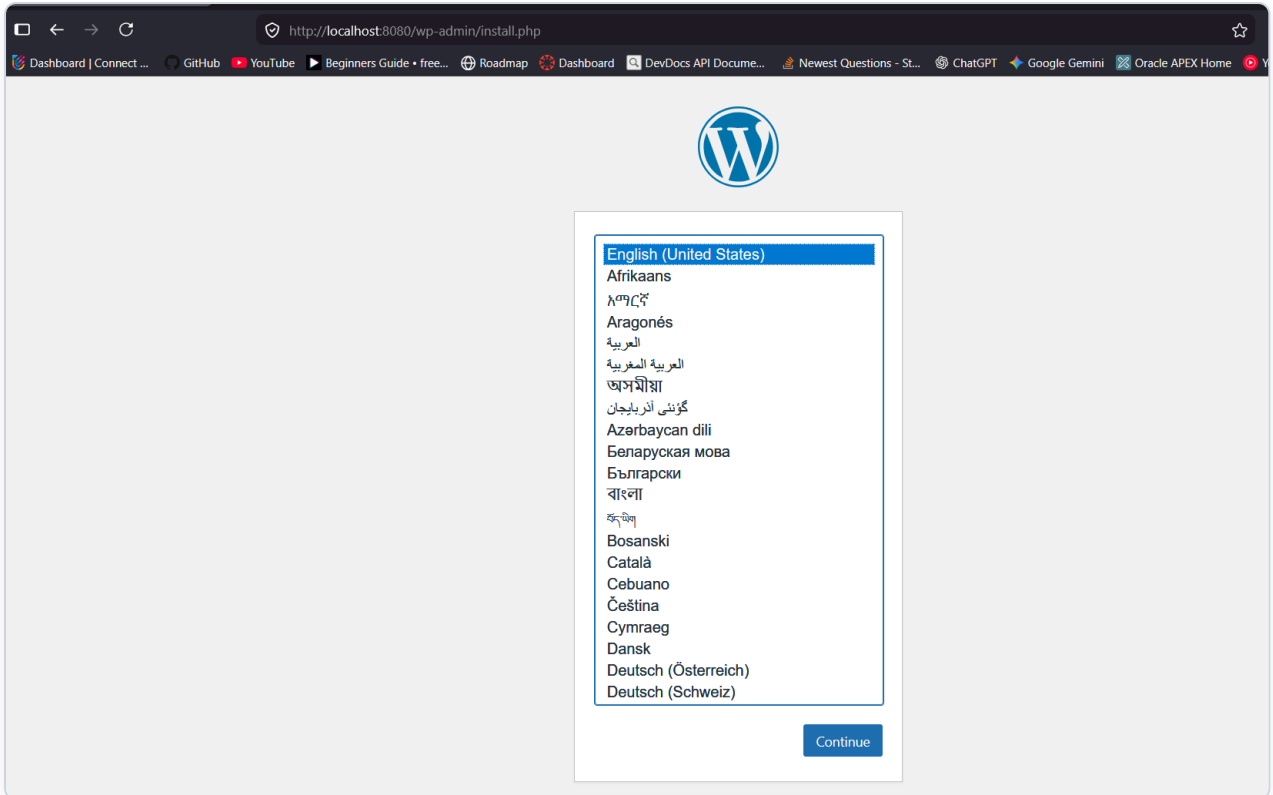
```
wpstack_wordpress.1.xxxx
wpstack_db.1.xxxx
```

Swarm now **manages containers automatically**.

Task 5 — Access the Application

Open browser:

```
http://localhost:8080
```



You should see the **WordPress setup page**.

Even though Swarm manages the containers, the application behaves the same.

Task 6 — Scale the Application

Scale WordPress service from **1 to 3 replicas**.

```
abhinav_pandey@LUCIF3R: ~/exp6$ docker service scale wpstack_wordpress=3
wpstack_wordpress scaled to 3
overall progress: 3 out of 3 tasks
1/3: running [=====>]
2/3: running [=====>]
3/3: running [=====>]
verify: Service wpstack_wordpress converged
```

Verify scaling:

```
abhinav_chaula@LUCIF3R:~/exp6$ docker service ls
```

ID	NAME	MODE	REPLICAS	IMAGE	PORTS
s0shrokwgbc1	wpstack_db	replicated	1/1	mysql:5.7	
t7f17sa6r0t5	wpstack_wordpress	replicated	3/3	wordpress:latest	*:8080->80/tcp

```
abhinav_pandey@LUCIF3R:~/exp6$ docker service ps wpstack_wordpress
```

ID	NAME	IMAGE	NODE	DESIRED STATE	CURRENT STATE	ERROR	PORTS
s88w74v0994n	wpstack_wordpress.1	wordpress:latest	LUCIF3R	Running	Running 7 minutes ago		
apw85cdg40or	wpstack_wordpress.2	wordpress:latest	LUCIF3R	Running	Running about a minute ago		
p2ct5h4opb1r	wpstack_wordpress.3	wordpress:latest	LUCIF3R	Running	Running about a minute ago		

```

abhinav chawla@LUCIF3R:~/exp6$ docker ps | grep wordpress
ff5ba9601355   wordpress:latest   "docker-entrypoint.s..."   About a minute ago   Up About a minute   80/tcp   wpstack_wordpress.3.p2
ct5h4opbl1rb8ygnv67n0zvf   wordpress:latest   "docker-entrypoint.s..."   About a minute ago   Up About a minute   80/tcp   wpstack_wordpress.2.ap
f8b6d250aad8   wordpress:latest   "docker-entrypoint.s..."   About a minute ago   Up About a minute   80/tcp   wpstack_wordpress.1.s8
w85cdg40or1b0b5lrcwepzl   wordpress:latest   "docker-entrypoint.s..."   7 minutes ago       Up 7 minutes       80/tcp   wpstack_wordpress.1.s8
8w74v0994ntwa4mtulwuvt

```

Now **three WordPress containers run simultaneously**.

Load Balancing

Even with multiple containers, users still access:

```
http://localhost:8080
```

Swarm creates an **internal load balancer** that distributes traffic across all containers automatically.

Task 7 — Test Self-Healing

Swarm automatically replaces failed containers.

Find a WordPress container:

```
docker ps | grep wordpress
```

Kill it:

```
docker kill <container-id>
```

Check service tasks:

```

abhinav_pandey@LUCIF3R:~/exp6$ docker service ps wpstack_wordpress
ID            NAME                IMAGE              NODE     DESIRED STATE   CURRENT STATE           ERROR   PORTS
s88w74v0994n wpstack_wordpress.1 wordpress:latest   LUCIF3R   Running         Running 11 minutes ago
apw85cdg40or wpstack_wordpress.2 wordpress:latest   LUCIF3R   Running         Running 5 minutes ago
p2ct5h4opblr  wpstack_wordpress.3 wordpress:latest   LUCIF3R   Running         Running 5 minutes ago

```

Swarm detects the failure and **creates a replacement container automatically**.

Task 8 — Remove the Stack

Clean up all services:

```
abhinav_pandey@LUCIF3R: ~/exp6$ docker stack rm wpstack
Removing service wpstack_db
Removing service wpstack_wordpress
Removing network wpstack_default
```

Verify removal:

```
abhinav_chamta@LUCIF3R: ~/exp6$ docker service ls
ID                NAME                MODE                REPLICAS            IMAGE                PORTS
```

Containers should be removed.

Compose vs Swarm

Feature	Docker Compose	Docker Swarm
Scope	single host	multi-node cluster
Scaling	basic	built-in service scaling
Load balancing	no	yes
Self-healing	no	yes
Rolling updates	no	yes
Service discovery	container names	DNS + VIP

Compose is best for development, while Swarm supports production orchestration.

Observations

1. The same Compose YAML file can be reused in Swarm.
 2. In Swarm, you manage **services instead of containers**.
 3. Swarm resolves port conflicts using an internal load balancer.
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Learning Outcome Questions

1. Why is Docker Compose not sufficient for production environments?
 2. What does `docker stack deploy` do differently from `docker compose up` ?
 3. How does Swarm provide self-healing?
 4. What happens if a container managed by Swarm is killed?
 5. Can the same Compose file be used for development and production?
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Conclusion

This experiment demonstrated container orchestration using Docker Swarm.

A multi-container WordPress application was deployed using a Compose file, scaled to multiple replicas, and tested for self-healing capabilities. Swarm provides built-in load balancing, automatic recovery, and cluster management, making it more suitable for production environment